

Cairo University

Faculty of Computers and Artificial Intelligence



CS251

Intro. to Software Engineering

Sigma

Software Requirements Specifications

Version 2.0

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CS251: Phase 2 – Sigma

Project: Toffee

Software Requirements Specifications

Contents

Team	3
Document Purpose and Audience	3
Purpose	3
Audience	3
Introduction	3
Software Purpose	3
Software Scope	3
Definitions, acronyms, and abbreviations	4
Requirements	5
Functional Requirements	5
Non Functional Requirements	8
System Models	10
Use Case Model	10
Enriched User Stories	11
System Navigation Map	41
Tools	42
Ownership Report	42



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Software Requirements Specifications

Team

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Document Purpose and Audience

Purpose

The main target of this document is to interpret program requirements that have been stated by the owners which is vital for cutting down cost and decreasing the time needed for progression.

Audience

- Software Development Team
- Owners

Introduction

Software Purpose

The purpose of Toffee Shopping System is to facilitate sweet orders for customers with many options to order and payment methods.

Software Scope

Toffee Shopping System helps users to buy sweets by package or kilograms with points, smart wallet or in delivery, and every item has a loyalty points scheme when users buy it.

It also allows administrators to view all orders, suspend a user, view statistics about daily and monthly sales, sales of each item over a specific period, most popular products, among other statistics.



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Definitions, acronyms, and abbreviations

Phrase	Definition
Smart Wallet	An electronic device, online service, or software program that allows one party to make an electronic transaction with another party bartering digital currency units for goods and services.
Performance	it can be defined as how efficiently a software can accomplish its tasks.
Compatibility	Compatibility defines how a system can co-exist with another system in the same environment.
Reliability	Reliability measures the ability of a piece of equipment to perform its intended function for a specific interval without failure.
Usability	The degree to which a software can be used by specified consumers to achieve quantified objectives with effectiveness, efficiency, and satisfaction in a quantified context of use.
Availability	Availability measures the ability of a piece of equipment to be operated if needed.



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Requirements

Functional Requirements

Requirement ID	Requirement Statement
FR01	Function that displays the store's catalog for the customer on the homepage of the site and makes some input fields; One for choosing specific category and one for searching for an item by name or by brand, it shells display table with all the items from the shop catalog and to buy items, s/he must register and login.
FR02	Function through which the user registers for the first time using an e-mail and its verification, the user's address, and a password that meets the stated requirements, such as not having less than 8 characters of uppercase and lowercase letters, numbers and signs. It shall display a box to input email , password and address if the user chooses to sign up. Registration is completed by sending an OTP to the registered email and the user has to confirm it to complete the registration; otherwise, registration is rejected.
FR03	Function through which the user logs in with email and password and then checks if that user exists in the database, it shall display a box to input username and password if the user chooses to log in.



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FR04	Function, the user can purchase with different payment methods such as in-delivery or through the smart wallet and others by showing the choices to the customer and making him choose freely with an introduction of the requirements of the good that he chose, which is required to complete the process such as his address or the credit card number and other requirements.
FR05	Function to make A logged-in user able to shop and add items to his or her shopping cart until s/he is finished, then s/he can check-out by making him able to select items by button and add it to the buy list and a button to put the numbers of units a user wants to add starting from a single unit up to 50 units from each item per order and a button for removing it. Items are sold by kilo to increase the number of kilos or decrease it.
FR06	Function to allow Shoppers to buy gifting vouchers to give as a gift to someone and every voucher has its unique code and can be redeemed once while making an order to reduce the total price by the value of the voucher by adding it to the list of items that are offered for customer to buy.
FR07	Function to allow the admin of the system to update the catalog with brand new items, cancel items or update item's info. An item has a name, category, description, image, brand, price, and a discount percentage (if any).



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FR08	Function to allow an admin to view all orders, set loyalty points scheme, suspend a user and view statistics. Statistics provide store owners with information whether the store is making profit and it includes daily and monthly sales, sales of each item over a specific period, most popular products, among other statistics.
FR09	Function to allow users to view their order history and they can reorder a previous order by clicking the "reorder" button which makes the same exact order with the same exact conditions.
FR10	Function to increase user's loyalty points. When making an order, a buyer earns loyalty points according to the scheme stated by the admin.
FR11	Function to take address as an input from user when he wants to make an order, shopper must specify the shipping address; if it is the same address registered on their profile or another address.
FR12	Function to check in case if the user is buying a gift for someone. If they are paying upon delivery, in this case they must enter a valid phone number. An OTP is sent to the phone number and the buyer must confirm it to verify the transaction.



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Software Requirements Specifications

FR13	Function that allows the user to: 1) Use one gift voucher or more to pay for their order or part of it. 2) Redeem some of their loyalty points to pay for the order or part of it. 3) Pay for the order or the remaining amount via payment methods like smart wallets. 4) Pay for the order or the remaining amount with cash upon delivery.
FR14	Function that makes an admins dedicated web app that allows them to monitor the system and do the assigned tasks.

Non Functional Requirements

- Non-functional requirements are requirements that specify criteria of the developed system. This can be used to judge operations of the system rather than specific behavior.

Measure	Details
Performance	● The registration or login screen must respond within 3 seconds. ● The system must check the money's availability in the smart wallet before accepting the order of the user.
Security	● Registration is completed by sending an OTP to the used email



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	and the user has to confirm it to complete the registration; ● Otherwise, registration is rejected.
Compatibility	● The program must work on different operating systems(Android, ios , browser.....etc).
Reliability	● The program should be reliable and provide catching of exceptions so that unintended results do not occur such as system crashes or data validation.
Usability	● The system should have a user -friendly interface.
Availability	● The system must work 24/7.

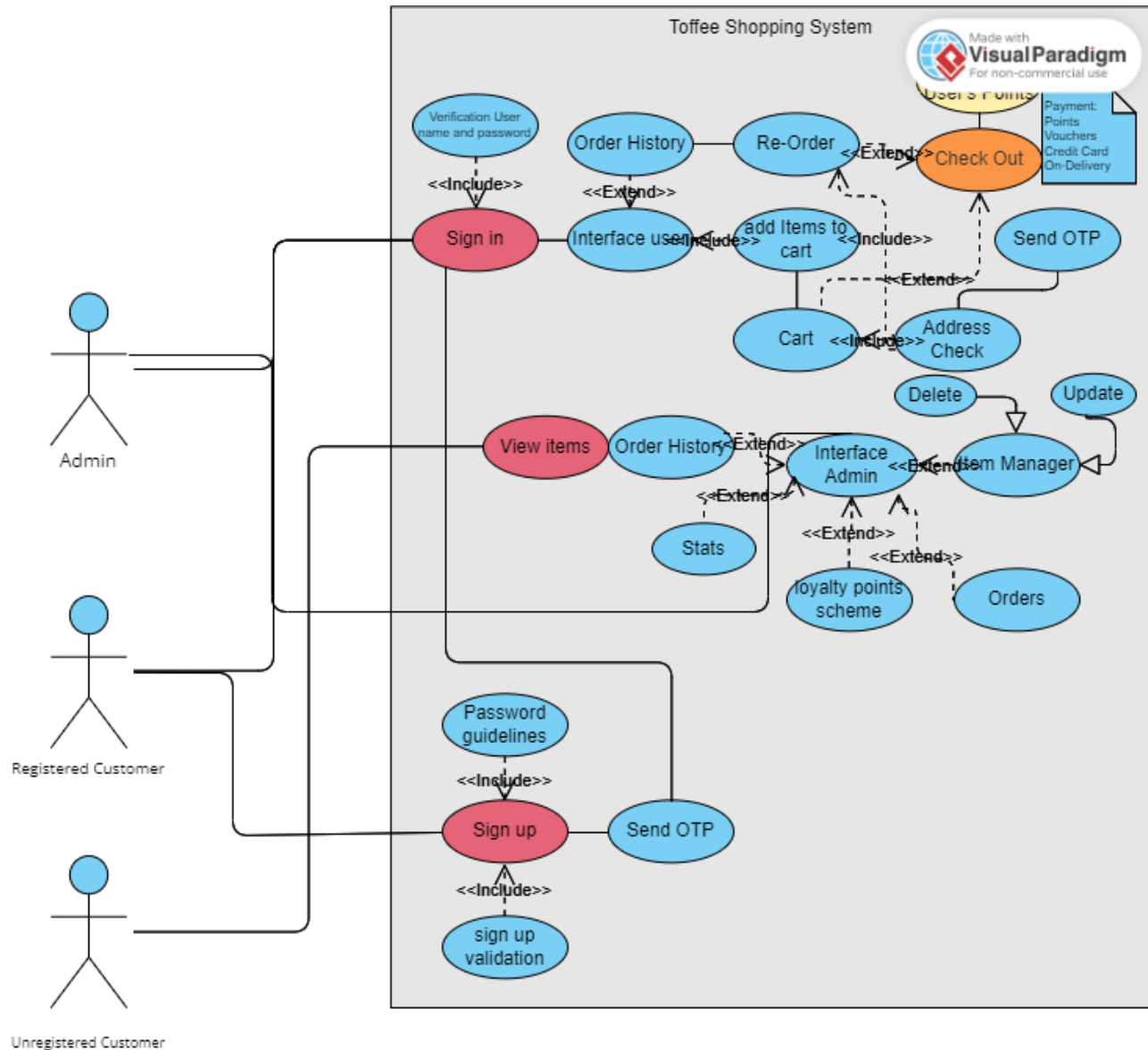


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Software Requirements Specifications

System Models

Use Case Model





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Software Requirements Specifications

Enriched User Stories

● User Story #1

User Story ID	User#1
User Story Name	Homepage without registration
Actors	Owner, User
Description	after user click on catalog button in the home page
Pre-condition	-
Post-condition	Catalog / sign in
Acceptance Criteria	Given I'm owner and I'm in the "Home Page" tab When I clicked on catalog to see it without sign in Then the system redirect user to catalog page

Scenarios

Normal Scenario

Actor	
Action	System Response
1-user click on catalog button	
	2- the system shows the catalog
OR 1-user clicks on sign in button	



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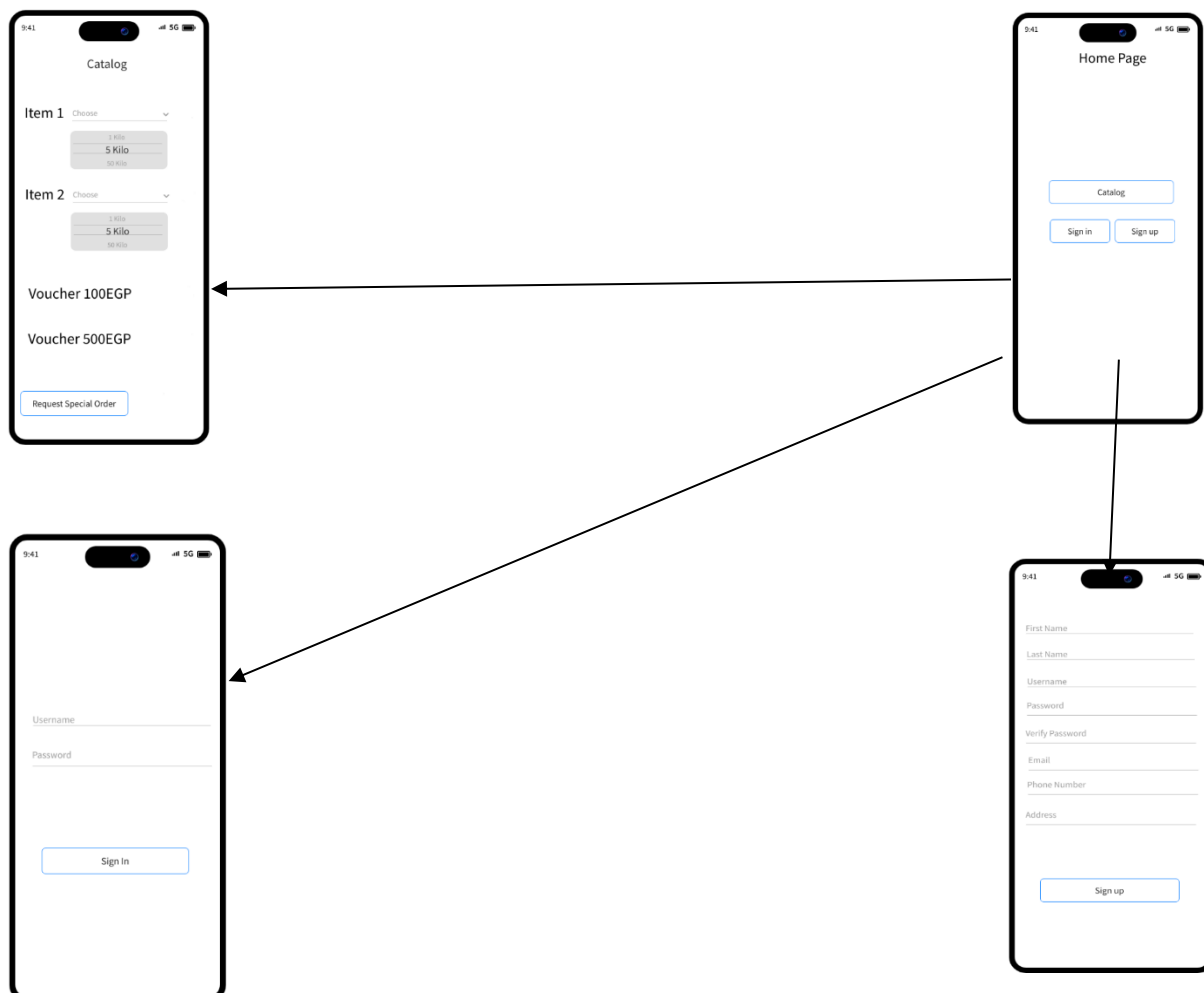
Software Requirements Specifications

	2- the system signed in
OR 1-user clicks on sign up button	
	2- the system signed up

Exceptional Scenario

there is not exceptional scenario

Screen Design





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• User Story #2

User Story ID	US#2
User Story Name	Sign in
Actors	Owner/ User
Description	As a playground owner/player I'd like to be able to log in to the application So that I can use its functionalities
Pre-condition	Home page without registration
Post-condition	Home page
Acceptance Criteria	Given I'm a logged-out system user and I'm on the Log-In page When I fill in the "Username" and "Password" fields with my authentication credentials and I click the Log-In button Then the system signs me in and redirect me to home page

Scenarios

Normal Scenario

Actor	
Action	System Response
1-User Clicks on log in	
	2- the System display sign in page 3- the system display box to the user to input email address and password



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Software Requirements Specifications

4- user fill the requirement data in the boxes	
	5- user verifies the user information
	6- the system redirect user to home page/admin homepage

Exceptional Scenario

Actor	
Action	System Response
1- User clicks on log in	
	2- System displays log in page
	3- System display boxes to input username and password
4- User fills in the boxes	
	5- System verifies the user information
	6- System finds input information invalid
	7- System displays “Please enter valid username/password.” message

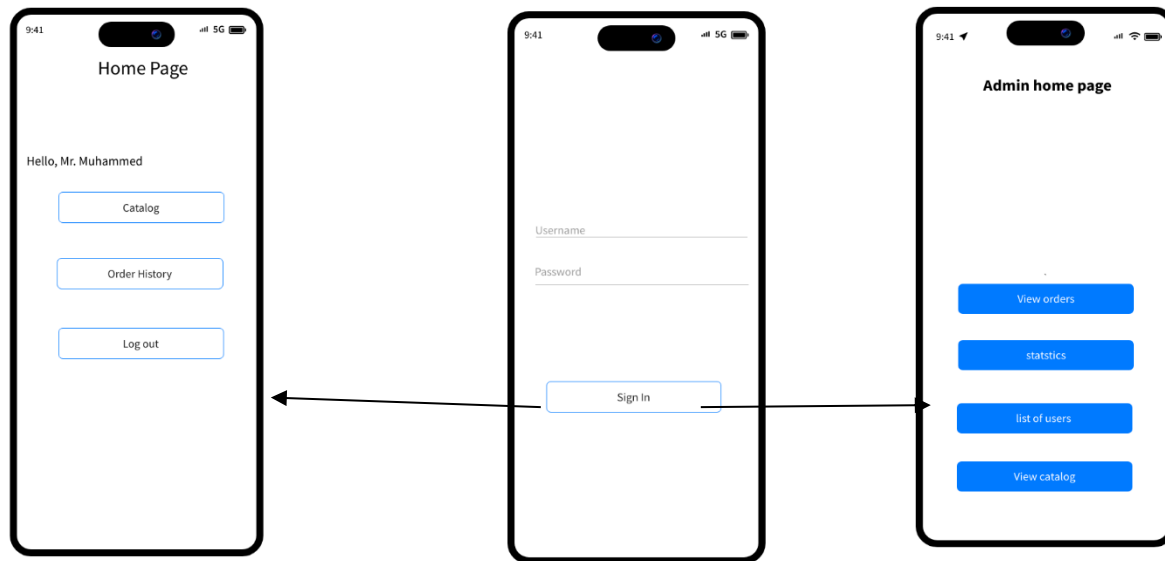


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Software Requirements Specifications

Screen Design



Data Dictionary:

Element Label	Type/Length	Data Validation / Business Rule
Email field	Text <100 characters	Structured Text
Password field	Text <100 characters	Structured Text

• User Story #3

User Story ID	US #3
User Story Name	Admin Homepage



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Software Requirements Specifications

Actors	admin/owner
Description	admin after login to the system can access four futures from four buttons
Pre-condition	login
Post-condition	system display page from four features pages
Acceptance Criteria	Given I'm a logged-in system admin and When admin click on stats button or list of users button or view catalog or view all orders button on the main page and Then the system redirects him to feature page

Scenarios

Normal Scenario

Actor	
Action	System Response
1-admin click on stats button	
	2-system display stats page
1-admin click on list of users button	
	2-system display list of users page
1-admin click on view catalog button	
	2-system display view catalog page
1-admin click on view all orders button	



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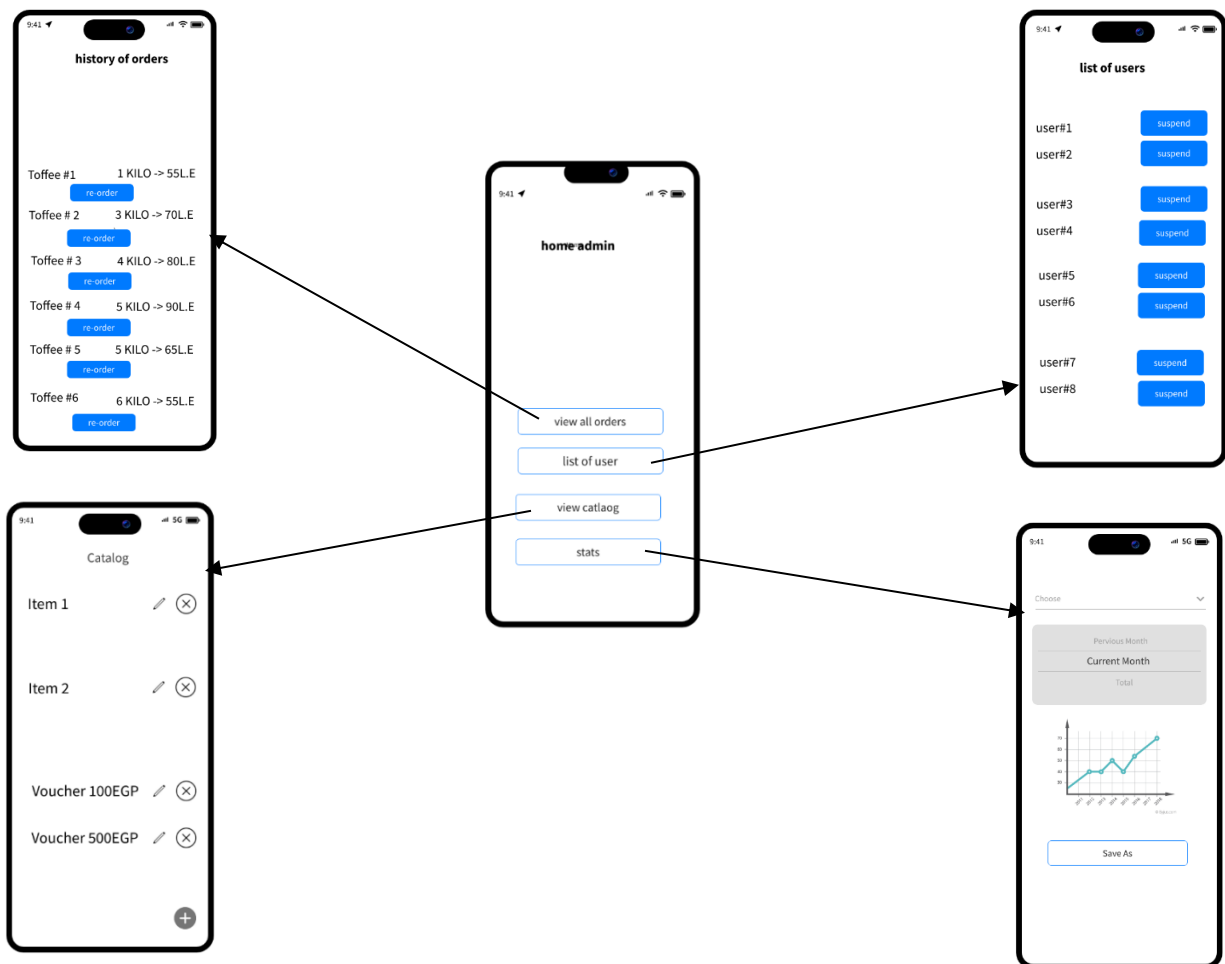
Software Requirements Specifications

	2-system display view all orders page
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Exceptional Scenario

No exceptional scenario for this user story.

Screen Design



• User Story #4



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Software Requirements Specifications

User Story ID	US #4
User Story Name	Add/Edit/Delete items in catalog
Actors	owner
Description	As a owner I'd like to be able to add/edit/delete item's catalog So that I can offer it for reservation
Pre-condition	Admin homepage
Post-condition	Edit items In catalog
Acceptance Criteria	Given I'm owner and I'm in the "Admin Home Page" tab When I fill in the required information for add / edit item. Then the system sends a request to add/edit my playground

Scenarios

Normal Scenario

Actor	
Action	System Response
1- User was on "Admin Home Page" and clicked on "Add/ Edit items Catalog".	



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Software Requirements Specifications

	2 - System displays several options which includes “Add item”, “Edit item” and “Delete item”
3- User clicks on add or edit item	
	4 - System displays several boxes for input, including name, category, description, image, brand, price, a discount percentage, and loyalty points scheme.
5- User fills the boxes	
6- User click to add or edit item	
	7- System update item’s info in database
8- if User click on delete item	
	9- System delete item from database

Exceptional Scenario

No exceptional scenario for this user story.

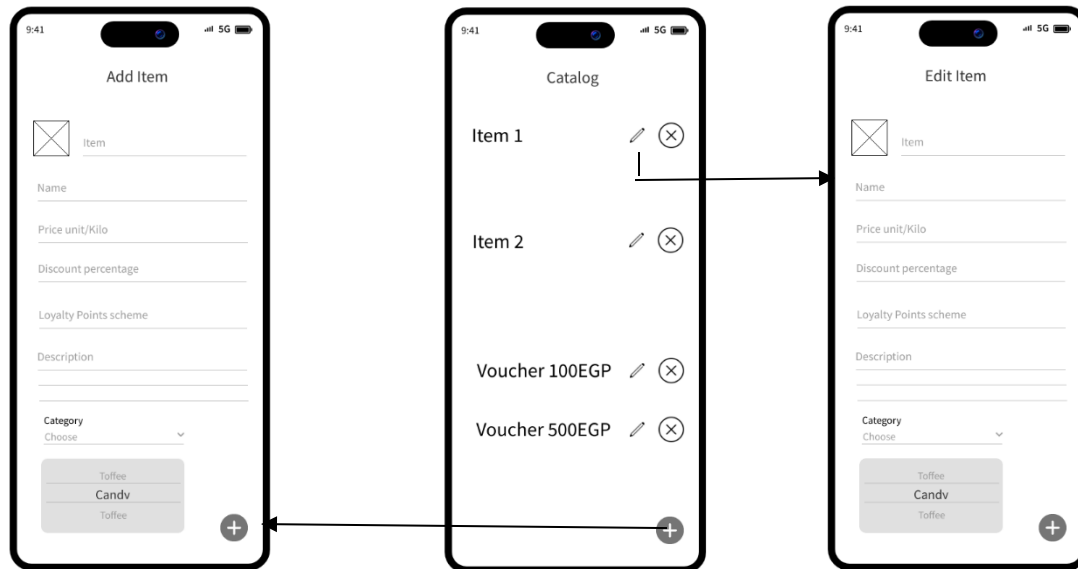


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Software Requirements Specifications

Screen Design



Data Dictionary:

Element Label	Type/Length	Data Validation / Business Rule
name field	Text < 30 Characters	Structured Text
category field	Text < 30 Characters	Select type
description field	Text < 1000 Characters	Structured Text
image field	.png file	image
brand field	Text < 30 Characters	Structured Text
price field	Integer	Integer > 0 and in EGP
discount percentage field	Integer <101	Integer > 0
loyalty points scheme	Integer	Integer > 0



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● User Story #5

User Story ID	US #5
User Story Name	Statistics Dashboard for Online Shop Owners
Actors	Owner
Description	As an online shop owner, I want to have a dashboard with statistics on the sales performance of my shop over a specific period so that I can make informed decisions to increase profits.
Pre-condition	Admin home page
Post-condition	-
Acceptance Criteria	Given I'm owner and I'm in the "Admin Home Page" tab. When I select type of statistics. Then the system display statistics.

Scenarios

Normal Scenario

Actor	
Action	System Response
1 - The shop owner logs in to their account and clicks on the "Statistics" tab	
	2 - The system displays a dashboard with an overview of the shop's sales performance for the



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Software Requirements Specifications

	current month, including total revenue, total number of orders, and most popular products.
3 - The system displays a dashboard with an overview of the shop's sales performance for the current month, including total revenue, total number of orders, and most popular products.	
4 -The shop owner can view the sales of each product over the selected period in a table or chart format.	
	5 - The system displays the customer data including average order value, lifetime value of customers, customer segments among your visitor traffic, etc in a separate tab.
6 - The shop owner can download the sales and customer data in XLS, CSV, or PDF format to further analyze the data in external tools such as Excel or Tableau.	

Exceptional Scenario

No exceptional scenario for this user story.

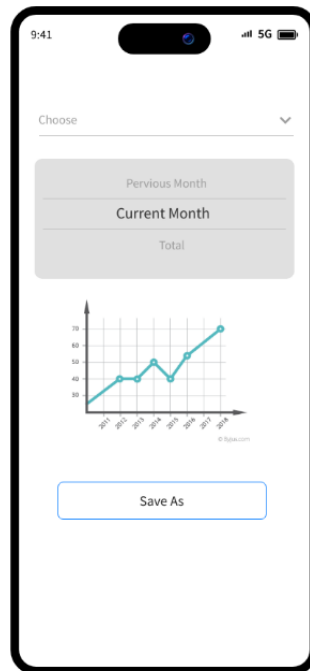


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Software Requirements Specifications

Screen Design



● User Story #6

User Story ID	US#6
User Story Name	View list of users
Actors	Admin
Description	Admin want to view list of users and all their information
Pre-condition	Admin homepage
Post-condition	Suspend user
Acceptance Criteria	the admin 's features to see the list of users



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Software Requirements Specifications

Scenarios

Normal Scenario

Actor	
Action	System Response
1- Admin click on list of users	
	2- The system shows the list of users.
3- Admin click on a random user	
	4- the system shows the list of the users

Exceptional Scenario

Actor	
Action	System Response
1- Admin click on list of users	
	2-The system refuse to respond

Screen Design





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Software Requirements Specifications

● User Story #7

User Story ID	US #7
User Story Name	view all orders
Actors	Owner
Description	view all the orders and all details
Pre-condition	Admin homepage
Post-condition	-
Acceptance Criteria	admin can see all the orders and all details of the orders and can suspend it

Scenarios

Normal Scenario

Actor	
Action	System Response
1 - Admin clicks on button view orders button	
	2- The System shows the list of the last orders

Exceptional Scenario

Actor	
Action	System Response
1 - Admin clicks on button view orders button	
	2- The System refuse to show the list of the last orders



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Software Requirements Specifications

Screen Design



• User Story #8

User Story ID	US #8
User Story Name	Sign-up
Actors	User
Description	As a User I'd like to be able to sign up to the app. So that I can sign in and access its services
Pre-condition	Homepage without registration
Post-condition	Homepage without registration
Acceptance Criteria	Given I'm a user and I'm on the sign-up. When I enter my credentials into the sign-up information fields and click the 'sign up' button



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Software Requirements Specifications

	Then the system will sign me u and redirect me to OTP page and save my data in Database
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Scenarios

Normal Scenario

Actor	
Action	System Response
1- User clicks on sign up	
	2 - System display Sign-up page
	3 - System display inputs boxes: first name, last name, User name, email, verify email, address, phone number and password
4 - User fills in the boxes	
	5 - System verifies the user information and Check the password guidelines in user's password
	6 - System send OTP to email
	7 - System display OTP Page
8 - User fill in the box	
	9 - System verify OTP
	10 - System accesses user to app functionalities and redirect user to homepage



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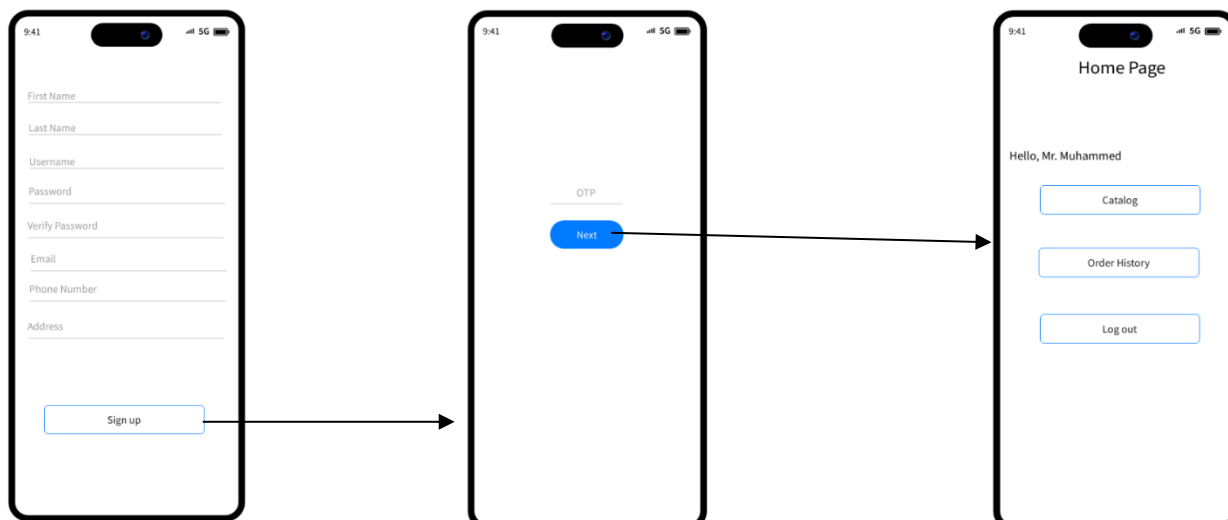
Project: Toffee

Software Requirements Specifications

Exceptional Scenario

Actor	
Action	System Response
1- User clicks on sign up	
	2 - System display log-in page
	3 - System display inputs boxes: email, verify email, address, password
4 - User fills in the boxes	
	5- System finds input information invalid
	6 - System displays “Please enter valid Information.” message

Screen Design





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Software Requirements Specifications

Data Dictionary:

Element Label	Type/Length	Data Validation / Business Rule
First Name and last name field	Text < 100 Characters	Structured Text
Location field	Text < 100 Characters	Structured Text
User ID field and valid ID	Integer < 15 digits	Integer > 0 and unique ID
Email field	Text < 100 Character	Structured Text and confirm via OTP in mail
Password field	Alphanumeric < 50 Characters	Password confirmation and must include an uppercase letter, numbers and/or special characters
Verify-password field	Alphanumeric < 50 Characters	Must be equal Password field
Phone number field	Integer < 15 digits	Integer > 0

● User Story #9

User Story ID	US #9
User Story Name	Homepage after login
Actors	user
Description	user after login can click on button from three buttons catalog, order history and log out
Pre-condition	login
Post-condition	catalog page/ history page /log out



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Project: Toffee

Software Requirements Specifications

Acceptance Criteria	Given I'm a logged-in system user and I'm on the main page. When user click on catalog then the system goes to catalog page or When user click on order history then the system goes to history page or When user click on logout then the system logout Then the system redirect me to selected page
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Scenarios

Normal Scenario

Actor	
Action	System Response
1-user click on catalog button	
	2- the system goes to catalog page
1-user click on order history button	
	2 - the system goes to history page
1-user click on logout button	
	2-the system redirect to "Homepage without sign in"



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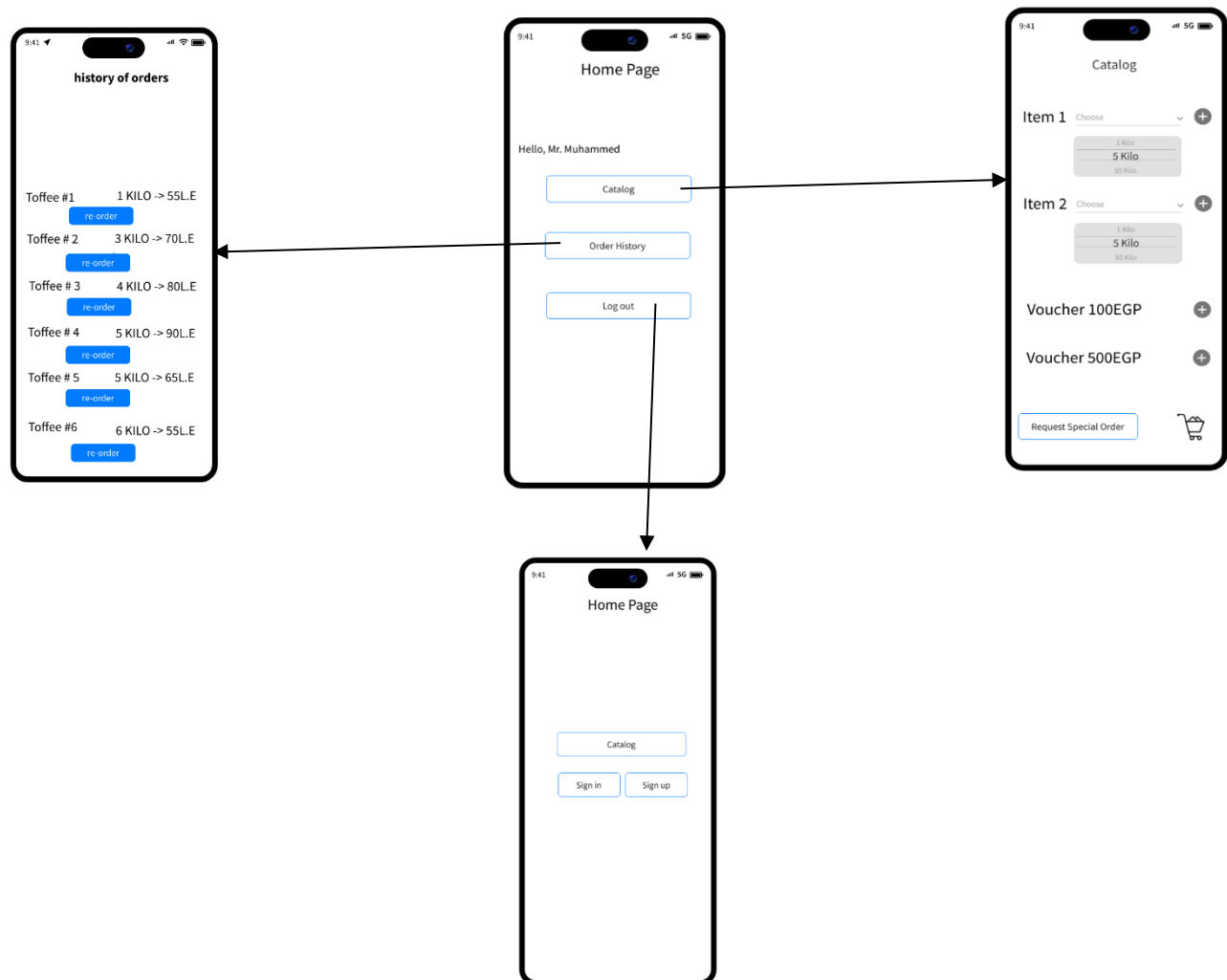
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Software Requirements Specifications

Exceptional Scenario

No exceptional scenario for this user story.

Screen Design



• User Story #10

User Story ID	US #10
User Story Name	Add items to cart



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Software Requirements Specifications

Actors	user
Description	user after login user can add items to cart
Pre-condition	Home Page
Post-condition	the cart page
Acceptance Criteria	Given I'm a logged-in system user and clicked on catalog button on the main page. When I selected an item from the catalog and clicked add to cart and clicked in cart button Then The system redirects me to cart page

Scenarios

Normal Scenario

Actor	
Action	System Response
1-User selects an item or more	
2-selects amount of it and clicks + Button	
	3 – The System selected item to cart
4 – User insert in button cart	
	5 – The system redirect user to cart page

Exceptional Scenario

No exceptional scenario for this user story.

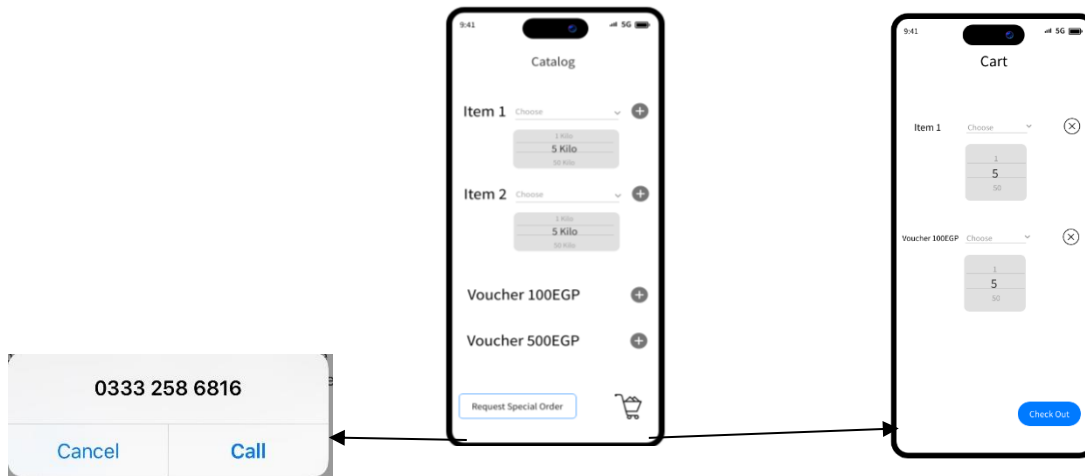


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Project: Toffee

Software Requirements Specifications

Screen Design



• User Story #11

User Story ID	US #11
User Story Name	reorder item from history
Actors	user
Description	user after login selects an item from the history page and decides to reorder it
Pre-condition	Homepage
Post-condition	Check out page
Acceptance Criteria	Given I'm a logged-in system user and clicked on history button on the main page and system goes to history page. When I selected an old order from the history page and clicked reorder

Scenarios



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Project: Toffee

Software Requirements Specifications

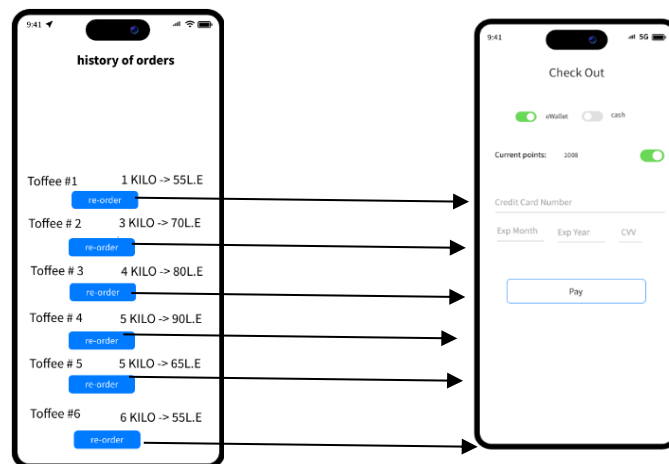
Normal Scenario

Actor	
Action	System Response
1-user click on history button from main page	
	2-system display history page
3-user select old order and click on reorder button	
	4-system display check out page

Exceptional Scenario

No exceptional scenario for this user story.

Screen Design



• User Story #12

User Story ID	US #12
User Story Name	cart page



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Software Requirements Specifications

Actors	user
Description	after user click on cart button in the main page user can check out by check out button or delete item from cart by delete button
Pre-condition	Catalog
Post-condition	Check out page
Acceptance Criteria	Given I'm a logged-in system user and clicked on cart button on catalog page. When I added an item from the catalog and clicked add to cart Then The system redirects me to check out page.

Scenarios

Normal Scenario

Actor	
Action	System Response
1-user click on check out button	
	2-system display check out page
1-user select item and click delete button	
	2-system delete item selected from cart

Exceptional Scenario

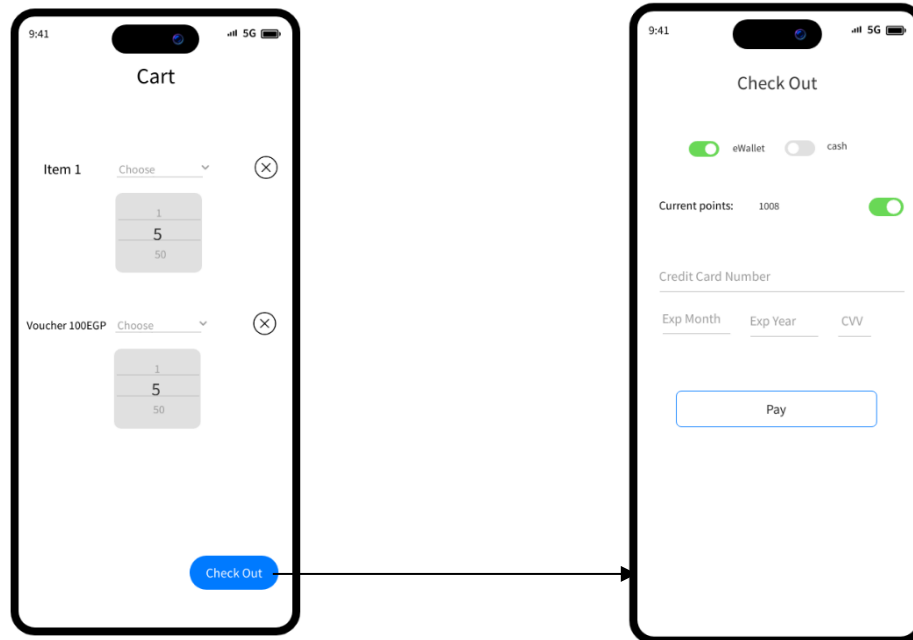


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Software Requirements Specifications

No exceptional scenario for this user story.

Screen Design



User Story #13

User Story ID	US#13
User Story Name	Check out
Actors	User
Description	As a User I'd like to check out and select the payment method and apply its information.
Pre-condition	Cart Page
Post-condition	Transaction page



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Software Requirements Specifications

Acceptance Criteria	Given I'm a logged-in system user and clicked on Checkout button and system redirected me to check out page. When I fill in the required information and click in Pay button Then the system redirects me to OTP Page and after verification redirect me to transaction page
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Scenarios

Normal Scenario

Actor	
Action	System Response
1- User selects an item to buy.	
	2- The system displays input boxes for the available payment methods (e.g., On cash or Smart Wallet payment).
3- User chooses which payment method to approach. IF the user selects Smart Wallet payment.	
	4- The system displays input boxes for Smart wallet information (e.g., Credit card holder name, Number, Expire date and password).
5- The user fills in the required boxes.	
	6- The system verifies the phone number and Email address.



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	The system verifies the user's Smart Wallet information.
7- if the user selects to cash payment	
	8- The system displays input boxes for personal information (e.g. registered address or a new address, Email address , Phone number)
9- The user fills in the required boxes.	
	10- If the information is verified, the system displays an input box for OTP to confirm the payment transaction.
11- The user fills in the required box.	
	12- The system verifies the OTP.
	13- The system displays an input box for the order's value the user wants to pay via Smart Wallet.
14- The user fills in the required box.	
	15- The system displays input boxes for whether the user wants to pay the remaining amount using loyalty points or on cash
16- The user fills in the required boxes.	
17- if the user selects to pay with loyalty points.	
	18- The system checks whether the user's loyalty points are enough to pay the remaining amount using it or not.
	19- The system discounts the required amount from the loyalty points.



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Software Requirements Specifications

	20- The system sets a new value to the user's remaining loyalty points.
	21- The system confirms the transaction and displays the checkout receipt.

Exceptional Scenario

Actor	
Action	System Response
1- User selects an item to buy.	
	2- The system displays input boxes for the available payment methods (e.g. On cash or Smart Wallet payment).
3- User chooses which payment method to approach. IF the user selects Smart Wallet payment.	
	4- The system displays input boxes for Smart wallet information (e.g., Credit card holder name, Number, Expire date and password).
5- The user fills in the required boxes.	
	6- The system finds error in eWallet
	7- The system displays “Error in payment method” message.

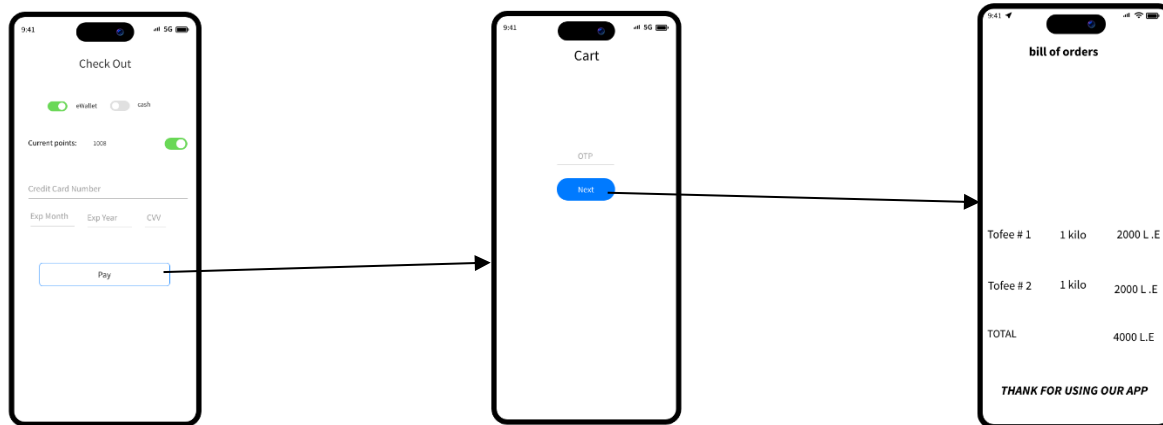


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Project: Toffee

Software Requirements Specifications

Screen Design



Data Dictionary:

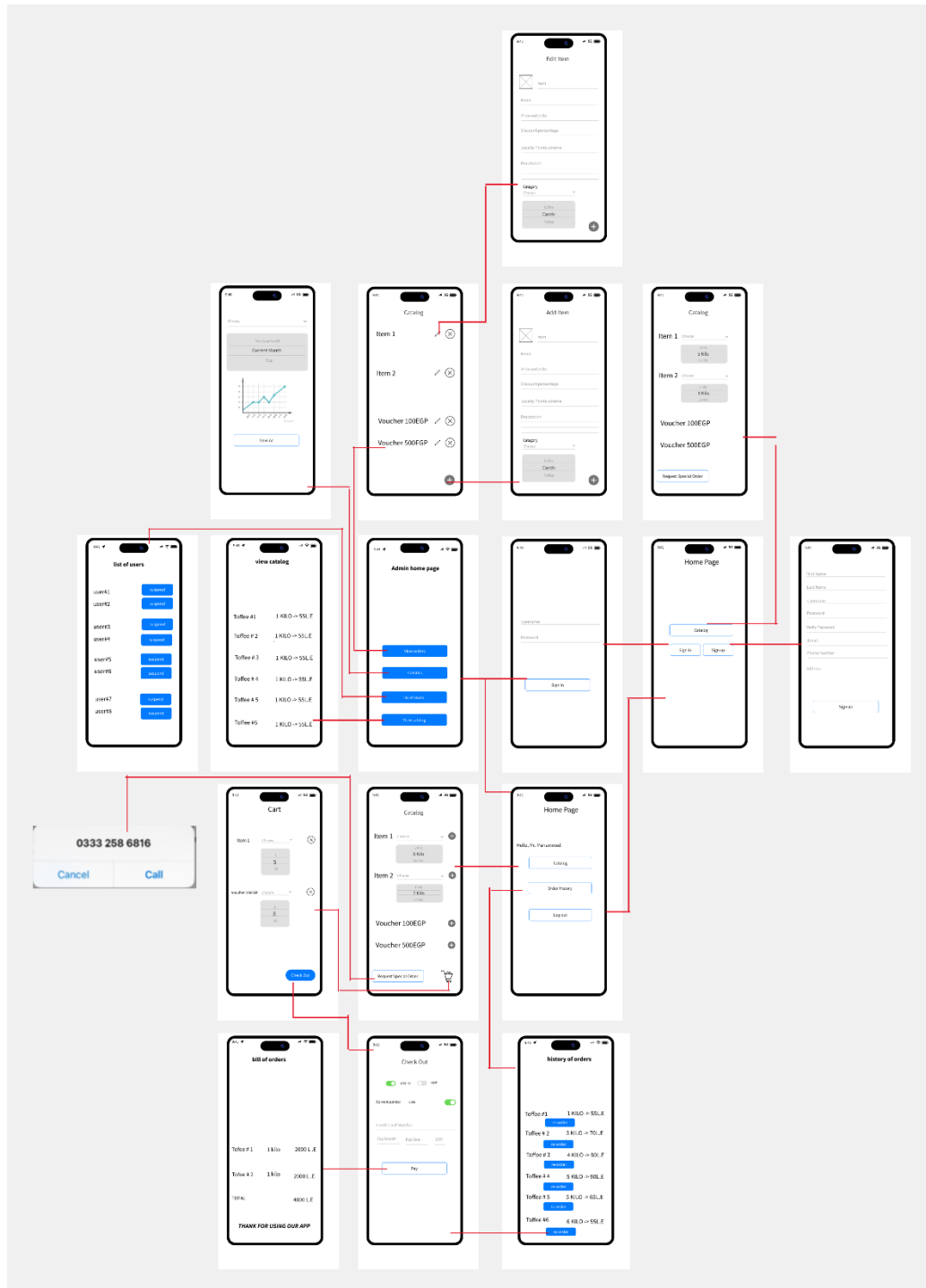
Element Label	Type/Length	Data Validation / Business Rule
Credit card number field	Integer = 16 digit	Credit card number
Exp Month	Integer < 13	Integer > 0, expire Month of credit card
Exp Year	Integer < 2040	Integer > 2022, expire year of credit card
CVV	Integer = 3 digits	Passcode of credit card



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Software Requirements Specifications

System Navigation Map





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Software Requirements Specifications

Tools

Visual Paradigm

Mockflow

Whiteboard microsoft

Ownership Report

Student	Items he created
Ayman Ashraf	Introduction User Stories
Muhammed Mahmoud	Non-functional Requirement System Navigation Map User Stories
Amr Abdelaziz	functional Requirement User Stories